

[LEDGER](#) · 2026-09-01 · POLYMER CHEMISTRY

WPUA-PAM@EG Organohydrogel Coatings via UV-Initiated Free-Radical Polymerization

REJECTED

This concept did **not** clear the framework.

The reasoning is published in full below.

What the assessment found

Incumbent benchmark

Bergolin 1-Component Topcoat 6P1600

Entry application

Leading Edge Protection (LEP) coatings for onshore wind turbine rotor blades

Measured advantage

322s droplet freezing delay at -30°C vs <20s for incumbent (>16x delay) and 39.8 kPa ice adhesion at -50°C vs >500 kPa for baseline (>12x reduction)

Time to first revenue

24 months

Gross margin at parity

46.4%

Startup CapEx

\$315,000

Payback from first sale

2.0 months

Capital productivity

18.29x

Regulatory risk

Moderate — qualification costs reported (\$450,000 estimated)

Why it failed

- Voided by: Mismatched comparison, Capital and time to revenue, Economics undisclosed, Entry application mismatch, Form-factor / handling regression
- Margin fails the operating-cost check

Assessment

The concept fails venture viability due to sub-70% gross margins (40.8%), excessive startup CapEx (\$315,000), and a 24-month time to first revenue. Severe operational regressions in handling toxic precursors and requiring field UV curing, alongside rain-erosion vulnerability in the entry application, void superiority.

Also flagged

- Evidence rests on non-peer-reviewed sources
- Duplicate references inflating the evidence base
- Price parity asserted, not sourced

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-09-01 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

biological or regulated processes that require appropriate expertise, facilities, permits and safety controls. Independently verify every figure, citation and regulatory requirement before committing capital.